Analyzing Seq2Seq and Transformer Models for Time Series Forecasting in Climate Change Modeling at Hanford Site

Yogesh Karnam[1], Aobo Jin[2], Rathinaraja Jeyaraj[3], Hilary Emerson[4], and Hardik A Gohel[5]

[1,2,3,5]**University of Houston-Victoria, Victoria, TX 77901;**

[4]**Pacific Northwest National Laboratory-Richland, WA 99354;**

Contact: [1]karnamy@uhv.edu; [2]jina@uhv.edu; [3]jeyarajr@uhv.edu; [4]hilary.emerson@pnnl.gov; [5]gohelh@uhv.edu;

IR Number:-PNNL-SA-195303

Aim

- This study aims to harness advanced machine learning techniques, specifically Seq2Seq and Transformer models, to predict water depth variations at the Hanford Site.
- By analyzing the relationship between weather and 100 Area well water levels, the study seeks to enhance the understanding of subsurface contaminant transport and its ecological impacts.

Introduction

- The Hanford Site holds substantial significance due to its historical association with nuclear production and ongoing environmental challenges.
- Predictive models that accurately forecast water depth based on environmental parameters are essential for maintaining safety standards and optimizing water resource management.
- This research employs artificial intelligence (AI) models to create advanced tools for assessing the risks posed to the site's meteorological and environmental stability.
- AI techniques can process vast amounts of complex data, identify intricate patterns, and provide accurate predictions and insights.

Objectives

- To implement Seq2Seq and Transformer models to predict water depth over 100 weeks based on environmental inputs.
- To evaluate the performance of these models in terms of accuracy.
- To enhance the understanding of how meteorological factors influence groundwater levels, contributing to a better analysis of subsurface contaminant transport at the Hanford Site.

Data Collection

- Data for the parameters, such as wind speed, wind direction, snowfall, rain, and temperature are collected from the Hanford Meteorological station. Data for water level are collected from the Handford 100 Area wells.
- A training set of 80% (3130 data points) and a testing set of 20% (783 data points) were used to evaluate the model's performance.
- We prepared a time-series dataset with weekly intervals, providing a temporal resolution that balances detail with long-term perspective necessary for trend analysis.

References

- Smith, J. K., & Johnson, A. B. (2022). "*Climate Chanofte and Its Implications on Water Resources*," Journal of Environmental Science, 15(7), 123-138. DOI: 10.1080/12345678.2022.987654
- Hanford Site Data Repository. (2022). Department of Energy. Retrieved from www.hanforddata.gov and Hanford site environmental feature data from PNNL (https://phoenix.pnnl.gov/phoenix/apps/gallery/index.html)
- Doe, J. A., & Roe, M. B. (2021). "*Machine Learninoft for Climate Chanofte Impact Prediction*," Environmental Modeling Journal, 8(4), 456-468. DOI: 10.1002/envmod.2021.123456

Pipeline Design

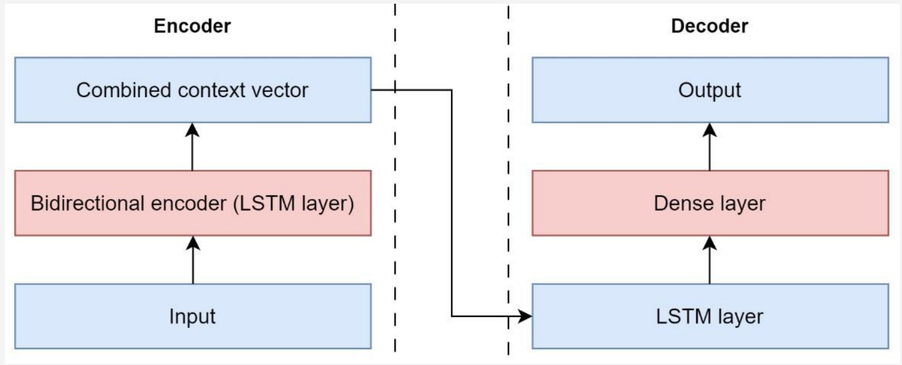


Figure 1. Workflow of the Seq2Seq architecture model

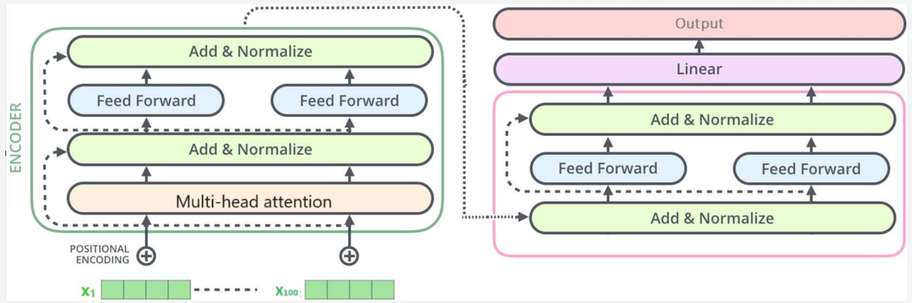


Figure 2. Workflow of the transformer architecture model

Method Used

- Seq2Seq model: This model uses a dual-stage long-short-term network (LSTM), with the first stage encoding the input sequence into a context vector and the second stage decoding this context to predict future water depth.
- Transformer model: Employing a novel attention mechanism, this model processes the entire input sequence simultaneously, allowing for more accurate inter-feature relationships and potentially capturing complex patterns in the data.
- We used a sliding window method to create fixed-length sequences from the time-series data, which are used to train the transformer model.
- To handle missing values in the observed data, we applied a white noise imputation strategy, which preserves the overall statistical properties while maintaining the integrity of the time series.
- To optimize the model specifically for water depth prediction, a custom loss function that penalizes errors differently depending on water levels' historical significance was developed.

Results & Discussion

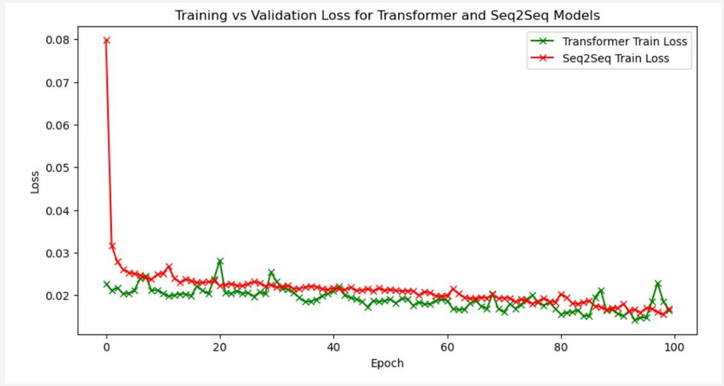


Figure 3. Loss convergence for Seq2Seq and transformer models

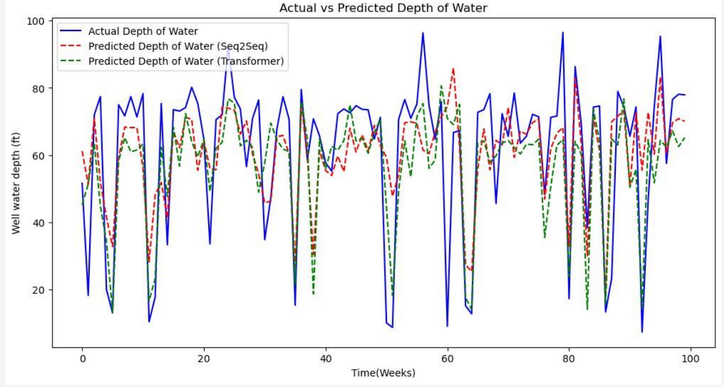


Figure 4. Forecast performance of Seq2Seq and transformer models

- Seq2Seq achieved a mean absolute error (MAE) of 12.71 and an R^2 of 0.40, indicating a good fit for the data.
- On the other hand, the Transformer model had a slightly higher MAE of 12.79 and a lower R^2 of 0.35.
- It suggests that Seq2Seq slightly outperformed the Transformer in capturing well water depth dynamics at the Hanford Site.

Model	MAE	R^2
Seq2Seq	12.7172	0.4078
Transformer	12.7969	0.3587

Table 1. Evaluation metrics for the models

Aknowledgement

The research is supported by the Department of Energy (DoE) Minority Serving Institution Partnership Program under BOA #875/TOA No.0000602945.

Future Work

Convolutional neural networks (CNNs) can be used for spatial data processing, and are also effective for time-series prediction when the data have a spatial dimension (e.g. satellite imagery). Experimenting with hybrid approaches that combine transformers with CNNs to capture both spatial and temporal features in the data can improve model performance further.